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DATA SET
DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

$$n = 300$$

Educational Level	Married	
	Yes	No
Bachelors or higher	187	113
Some college	162	138
High school or less	149	151
	498	402

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0 : Marriage rates are the same for all educational levels

H_a : Marriage rates are not all the same.

Educational level	Yes	No
Bachelor or higher	166	134
Some College	166	134
High School or less	166	134

$$\begin{aligned} \chi^2 &= \sum \frac{(O-E)^2}{E} \\ &= \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(149-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134} \\ &= 10.06 \end{aligned}$$

$$df = 2$$

$$\chi^2_{0.01, 2} = 9.210$$

$$\therefore 10.06 > 9.210$$

$\therefore$ Reject H_0

$$\therefore \text{Bachelor's or higher} : \frac{187}{300} = 0.623 \quad \text{Some college} : \frac{162}{300} = 0.540 \quad \text{High school or less} : \frac{149}{300} = 0.497$$

$\therefore$ adults with higher educational level have higher marriage rates.

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence		
	Apartment	Duplex	Single Residence
1	8	20	1
2	16	8	9
3	10	10	14
4 or More	6	2	16
	40	40	40

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H_0 : The family size distributions are the same for the three types of residences

H_a : The family size distributions are not the same for the three types of residences

Family members	Apartment	Duplex	Single Residence
1	9.67	9.67	9.67
2	11	11	11
3	11.33	11.33	11.33
4 or More	8	8	8

$$\begin{aligned} \chi^2 &= \frac{(O-E)^2}{E} = \frac{(8-9.67)^2}{9.67} + \frac{(20-9.67)^2}{9.67} + \frac{(1-9.67)^2}{9.67} + \frac{(16-11)^2}{11} + \frac{(8-11)^2}{11} + \frac{(9-11)^2}{11} \\ &\quad + \frac{(10-11.33)^2}{11.33} + \frac{(10-11.33)^2}{11.33} + \frac{(14-11.33)^2}{11.33} \\ &\quad + \frac{(6-8)^2}{8} + \frac{(2-8)^2}{8} + \frac{(16-8)^2}{8} \\ &= 36.50 \end{aligned}$$

$$df = 6$$

$$\chi^2_{0.01, 6} = 16.812$$

$$\therefore 36.5 > 16.812$$

$\therefore$ Reject H_0 , there is sufficient evidence.

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

- Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

a) H_0 : Above & Below have the same distribution
 H_a : Below town is lower than above town. ($A > B$)

Value	4.7	4.8	4.8	4.9	4.9	4.9	5.0	5.0	5.1	5.2
Group	B	A	B	A	B	B	A	B	A	A
Rank	1	2.5	2.5	5	5	5	7.5	7.5	9	10

$$R_A = 2.5 + 5 + 7.5 + 9 + 10 = 34$$

$$R_B = 1 + 2.5 + 5 + 5 + 7.5 = 21$$

$$n_1 = 5 \quad n_2 = 5 \quad \alpha = 0.05$$

$$T_L = 19$$

$$\therefore 21 > 19, \nless 12$$

$$\therefore \text{Non-reject } H_0.$$

b) $H_0: \mu_A = \mu_B$
 $H_a: \mu_A > \mu_B$

$$\bar{X}_A = 5$$

$$\bar{X}_B = 4.86$$

$$\bar{X}_A - \bar{X}_B = 0.14$$

$$s^2_A = 0.025$$

$$s^2_B = 0.013$$

$$s^2_P = \frac{4(0.025) + 4(0.013)}{8}$$

$$= 0.019$$

$$s_P = 0.1378$$

$$t = \frac{5 - 4.86}{0.1378 \sqrt{\frac{1}{5} + \frac{1}{5}}}$$

$$= 1.61$$

$$df = 5 + 5 - 2$$

$$= 8$$

$$t_{0.05, 8} = 1.86$$

$$\therefore 1.61 < 1.86$$

$$\therefore \text{Non-reject } H_0$$

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

Deaf Children	Hearing Children
2.75 (15)	.89 (1)
2.14 (11)	1.43 (7)
3.23 (18)	1.06 (4)
2.07 (10)	1.01 (3)
2.49 (14)	.94 (2)
2.18 (12)	1.79 (8)
3.16 (17)	1.12 (5.5)
2.93 (16)	2.01 (9)
2.20 (13)	1.12 (5.5)
Rank Sum	126
	45

H_0 : The distributions are same.

H_a : The distributions are different.

$$T = 45$$

$$n_1 = 9, n_2 = 9, \alpha = 0.05$$

$$T_L = 62$$

$$RR: T \leq 62$$

$$\therefore T = 45 \leq 62$$

$$\therefore \text{Reject } H_0$$

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

	+	+	+	Pair	+	+	+
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

$$+ = 6 \quad - = 1 \quad \text{no '0'} \Rightarrow n = 7$$

$$H_0: p = 0.5$$

$$H_a: p \neq 0.5$$

$$X \sim B(7, 0.5) \quad , \quad \alpha \leq 0.10$$

$$P(X \leq 0) = 0.5^7 = 0.0078$$

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$= 0.0078 + 7 \times 0.5^7$$

$$= 0.0625$$

$$P(X \leq 1) = 0.0625 > 0.05 \quad (\text{Two-tailed} : 2(0.0625) = 0.125 > 0.10)$$

$$X = 0 \quad \text{or} \quad X = 7$$

$$\alpha = P(X = 0) + P(X = 7)$$

$$= 0.0078 \times 2$$

$$= 0.0156 < 0.10$$

$$X = 6 \notin R$$

$$\therefore \text{Non-reject } H_0$$

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	14	3	3
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
7	9	9	18	4	6
8	7	8	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

- Use the binomial tables in Appendix I to find the exact rejection region for the test.
- Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- Compare the results of parts a and b.

$$+ = 11 \quad - = 9 \quad \Rightarrow n = 20$$

$$H_0: p = 0.5$$

$$H_a: p \neq 0.5$$

$$a) X \sim (20, 0.5)$$

$$P(X \leq 5) = 0.0207 \quad \geq (0.0207) = 0.0414$$

$$P(X \leq 6) = 0.0577 \quad \geq (0.0577) = 0.1154$$

$$X \leq 5 \text{ or } X \geq 15$$

$$T = \min(+, -)$$

$$T = \min(11, 9) = 9$$

$$\text{DR: } T \leq 5$$

$$\because 9 > 5$$

$$\therefore \text{Non-reject } H_0$$

$$b) Z = \frac{X - \frac{n}{2}}{\sqrt{\frac{n}{4}}}$$

$$X = 11, n = 20$$

$$Z = \frac{11 - \frac{20}{2}}{\sqrt{\frac{20}{4}}}$$

$$= 0.45$$

$$\text{two-tailed, } \alpha = 0.025$$

$$Z_{0.025} = 1.96$$

$$\text{RR: } Z < -1.96 \text{ or } Z > 1.96$$

$$\because 0.45 < 1.96$$

$$\therefore \text{Non-reject } H_0$$

$$c) (a) \& (b) \text{ have the same result}$$

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

- a. Test using the sign test. Choose your rejection region so that α is near .05.
- b. Why might it be inappropriate to use the Student's t -test in analyzing the data?

Drug A			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Drug B			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4 -
2	83	69	83.1 -
3	91	73	80.2 +
4	47	35	74.5 -
5	60	42	70.0 -
6	27	22	81.5 +
7	69	52	75.4 -
8	72	57	79.2 -
9	89	76	85.4 -
10	46	37	80.4 -

$$a) + = 2 \quad - = 8 \quad n = 10$$

$$H_0: p = 0.5$$

$$H_a: p \neq 0.5$$

$$T = \min(+, -)$$

$$T = \min(2, 8) = 2$$

$$X \sim B(10, 0.5)$$

$$P(X \leq 1) = P(X=0) + P(X=1) \\ = \frac{1 + 10}{2^{10}} \\ = 0.0107$$

$$2(0.0107) = 0.0214$$

$$RR: X \leq 1 \text{ or } X \geq 9$$

$$RR: T \leq 1$$

$$\therefore T=2$$

$$\therefore \text{Non-reject } H_0$$

b) The data are percentages & may not be normally distributed, so the sign test is more appropriate.